

PRIYANKA ASTHANA

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PROFESSIONAL SUMMARY

Final-year BTech CS student graduating in 2026 with a published patent, two Springer Nature submissions, and hands-on experience building RAG pipelines, constrained LLM systems, and multi-agent architectures. Research sits at the intersection of NLP and real-world deployment. ERA AI Fellowship Finalist, Cambridge University.

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University (AKTU)

2022 – 2026

BTech (Hons.) Computer Science & Engineering | Minor: Robotics | CGPA: 8.63/10 | I-DIV with DISTINCTION | Gold Medalist

Ashoka Institute of Technology and Management, Varanasi

Relevant Coursework: Algorithms, Data Structures, Linear Algebra, Probability & Statistics, Optimization Methods, Discrete Mathematics, Theory of Computation

RESEARCH & INTELLECTUAL PROPERTY

Published Patent - AI-Driven Conversational Medical Voice Assistant [↗](#)

Mar 2026

Indian Patent Application No. 202611012869

- Designed a multi-agent conversational system routing patient symptom inputs to specialist agents (cardiologist, dermatologist, etc.) using NLP-based intent classification and semantic parsing.
- Stress-tested across 500 simulated patient scenarios, achieving 92% disease detection accuracy.

Embedding Model Selection for Domain-Specific RAG:

2026

Zenodo. <https://doi.org/10.5281/zenodo.20399130> | SN Computer Science (Springer Nature)

- Conducted systematic benchmark evaluation of three embedding models (MiniLM, MPNet, multilingual MiniLM) on a domain-specific Indian cultural heritage corpus using RAGAS metrics: faithfulness, answer relevancy, and factual correctness.

Priority-Weighted Spatial Framework for Emergency Blood Allocation:

2026

Zenodo. <https://doi.org/10.5281/zenodo.18149564> | Journal of Computational Social Science (Springer Nature)

- Developed and published a novel GeoAI framework combining spatial clustering with time-bound token reservation; designed for real-world emergency resource allocation under scarcity conditions.

EXPERIENCE

IIT BHU | NLP Engineer Intern

Feb 2026 – Jul 2026

- Built NLP-driven test automation pipeline for heritage text systems, processing 1,000+ text inputs for semantic accuracy validation - directly supporting digital preservation of historical corpora.
- Diagnosed and resolved 20+ NLP parsing failures in live systems, improving pipeline reliability by 25%; documented root cause analysis and recommended architectural fixes to the senior engineering team.

Hindalco Industries, Aditya Birla Group | Business Analyst Intern

Jul 2025 – Sept 2025

- Built SQL queries against Oracle ERP systems to extract, validate, and transform data; identified process inefficiencies through gap analysis and translated findings into Tableau dashboards for senior leadership.
- Communicated performance metrics and analytical insights directly to senior team members, reducing manual processing time by 25% and improving API query performance by 10%.

Bharat Space Education Research Centre | AI Research Intern

Dec 2025 – Feb 2026

- Applied ML algorithms to real-time sensor data fusion across 5 experimental UAS modules in simulated space mission environments, contributing to trajectory prediction and subsystem latency reduction of 20%.

PROJECTS

Kashivani | Domain-Specific RAG Pipeline for Cultural Heritage [↗](#)

Sep 2025 – Mar 2026

- Designed and built an end-to-end RAG pipeline for Varanasi cultural heritage knowledge, covering ingestion, chunking, embedding, FAISS-based retrieval, and LLM-powered generation.
- Conducted rigorous embedding model evaluation using RAGAS framework (faithfulness, answer relevancy, factual correctness) — reducing hallucinations by 40% through systematic model selection.
- Research published on Zenodo and submitted to Springer Nature; demonstrates ability to translate domain expertise into production-ready AI solution design with measurable quality benchmarks.

DiagnoseAI | Multi-Agent Medical Symptom Assessment System [↗](#)

Apr 2025 – Sep 2025

- Architected a voice-first multi-agent system integrating speech-to-text, NLP-based symptom routing, specialist LLM agents (cardiologist, dermatologist, etc.), and structured report generation.
- Built agent orchestration and routing logic using LangChain, enabling structured handoffs between specialist LLM agents and consistent context management across the conversational pipeline.
- Achieved 92% disease detection accuracy across 500 simulated patient scenarios; work patented.

RAKT | Tokenized GeoAI Emergency Blood Allocation Framework [↗](#)

Dec 2024 – May 2025

- Engineered a spatial-temporal allocation system using K-means clustering on GPS coordinates with haversine distance, applied to real inventory data from e-Raktkosh (India's national blood transfusion portal).
- Implemented time-bound token reservation mechanism to prevent concurrent allocation conflicts; solving a real coordination problem in emergency logistics.
- Achieved 100% high-priority allocation and 6.6% distance reduction under simulated scarcity conditions; framework is directly generalizable to enterprise resource allocation problems.

TECHNICAL SKILLS

- **AI/GenAI & RAG:** RAG pipeline design, embedding model evaluation, constrained LLM reasoning, multi-agent architectures, LangChain, Agentic AI workflows, prompt engineering, RAGAS framework, FAISS
- **ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Sentence Transformers, HuggingFace
- **Data & Analytics:** Pandas, NumPy, Tableau, Oracle ERP data pipelines, spatial data analysis (K-means, haversine), SQL
- **Programming:** Python (primary), JavaScript, React, TailwindCSS, Git
- **Cloud & AI Platforms:** Microsoft Azure (AI-900, AI-102 certified), Oracle AI Vector Search (certified)

CERTIFICATIONS

- **Microsoft Azure AI-900** - AI Fundamentals
- **Microsoft Azure AI-102** - Designing and Implementing AI Solutions (Azure OpenAI, Azure AI Search, Responsible AI)
- **Oracle AI Vector Search Professional Certification** - Applied AI Systems
- **Oracle Cloud Infrastructure Certification** - Generative AI Professional

AWARDS & HONOURS

ERA:AI Fellowship Finalist, Technical Track | Cambridge University, UK

Apr 2026

- Selected through a competitive multi-stage global application for a fully funded AI safety research fellowship at Cambridge; one of the few finalists shortlisted in the technical track from a worldwide applicant pool.

NPTEL STAR Award | IIT Kanpur

Aug 2025

- Awarded to top-performing students nationally across NPTEL courses; recognises exceptional academic performance evaluated by IIT Kanpur faculty.

LEADERSHIP

- Student Council President | AITM, Varanasi
- Patriotic Youth Ambassador | Veterans India

May 2025 - May 2026

Aug 2025 - Sept 2025